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Banking on Connectivity: Internet Exposure and Women's Financial Autonomy

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Abstract

We examine whether women's exposure to the internet enhances their economic agency in India. Using nationally representative data and an instrumental variables strategy that exploits plausibly exogenous variation in district-level mobile tower density, we identify the causal effect of internet access on women's financial control and use of formal banks. We find that internet exposure increases women's overall financial autonomy by 10 percentage points. We document improvements in independent mobility, employment, and financial awareness which we consider potential mechanisms of our estimated effects. Disaggregated outcomes show higher likelihood that women have money of their own over which they full control, as well as increased ownership and active use of a bank account. Our results are robust to additional controls and a battery of sensitivity analyses. Heterogeneity analyses indicate that effects are concentrated among women with at least secondary education, and among those from historically disadvantaged social groups. Overall, our findings highlight the importance of digital connectivity in enhancing the usage of financial resources, with important implications for women's autonomy and participation in the formal economy.

JEL Classifications: D13, G20, I31, J16, O12

Keywords: Internet Exposure, Mobile Tower Density, Financial Autonomy, Instrumental Variable, India

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1 Introduction

Financial inclusion is widely recognized as a crucial driver of women’s empowerment, as it empowers them to control their financial resources, facilitate savings, and enhance their ability to manage risks and invest in their well-being (Swamy, 2014; Yang et al., 2022). While financial inclusion has gained momentum in contemporary development policy, much of the discourse still centres on women’s access to such products and services. However, effective and sustained use of financial services requires more than mere access. Rather, it depends on women’s autonomy to utilize these products and services. This is especially important in patriarchal contexts where women face layered constraints to using bank accounts, which can range from limited awareness and literacy to restrictions on mobility and decision-making. This paper asks whether access to the internet can expand women’s financial autonomy, both in terms of owning an account and actively using it, as well as in terms of having complete control of and independent decision-making about their own money. We examine this question in the context of India using nationally representative data.

Women’s financial inclusion is recognized as a critical aspect of empowerment and plays an important role in human development (Doepke & Tertilt, 2019; Orso & Fabrizi, 2016; Zelu et al., 2024). Access to mobile money has been shown to increase per capita consumption of households and reduce poverty incidence, especially in households that are headed by women (Suri & Jack, 2016). Bank expansion has also been linked to better health outcomes, including reductions in non-chronic diseases, and particularly benefits women with access to safe pregnancy services (Cramer, 2025). Financial inclusion is also said to have caused a reduction in dowry-related deaths and domestic cruelty (Bhukta et al., 2025). That said, Shreemoyee et al. (2025) points to an increase in IPV due to women’s greater financial inclusion, often interpreted as a response to shifts in traditional patriarchal norms. Financial inclusion has also been shown to have positive impacts on women’s labour force participation (Zelu et al., 2024), entrepreneurial opportunities (Yang et al., 2022), and income levels (Swamy, 2014). Taken together, existing evidence suggests that expanding women’s financial inclusion and autonomy has consequences not only for their own well-being but also for broader patterns of social and economic development.

However, notwithstanding the increase in the number of bank accounts in India, women still own only 39.2 per cent of all bank accounts and contribute to just 39.7 per cent of total deposits (Ministry of Statistics and Programme Implementation (MoSPI), 2025). Moreover, roughly only 30% of women in India utilize Unified Payments Interface (UPI) for financial transactions (Gaur, 2023). This shows that massive scope for improvement remains in achieving true financial inclusion. One of the primary reasons for lower financial inclusion among women is attributed to the digital divide as well as the lack of financial freedom (Basu Chandola, 2025). Therefore, in our analysis, we move beyond financial inclusion to focus on financial autonomy, which includes not only access to but also use of financial services, as well as decision-making over one’s own finances, in order to provide a more holistic perspective.

Our empirical strategy exploits plausibly exogenous variation in India’s mobile internet infrastructure to estimate the causal impact of digital exposure on women’s financial autonomy. We argue that internet access can improve women’s access to information and reduce bargaining frictions within the household

by increasing their awareness about government transfers, savings instruments, and mobile banking tools, and by strengthening women’s intra household bargaining position. In doing so, we speak to the gap between having and using a bank account, with particular relevance for women who are otherwise marginalized from the formal financial system.

India’s digital transformation provides a natural setting for this inquiry. Early internet access was confined to research institutions and government networks such as Education and Research Network (ERNET) and National Informatics Centre Network (NICNET), with commercial availability arriving only in the mid-1990s via the state-owned Videsh Sanchar Nigam Limited (VSNL) (Vinayak, 2020). Despite private entry in the late 1990s, adoption remained slow due to high data costs, limited bandwidth, and sparse infrastructure. By 2008, the country counted roughly 11 million subscribers, a relatively modest figure given the country’s population size. (Mangla & Singh, 2021).

Momentum accelerated in the 2010s, especially after the launch of Digital India in 2015 to bolster infrastructure and rural connectivity (TOI Business Desk, 2025). A decisive market shock arrived in 2016 with Reliance Jio’s low-cost 4G data and free voice services, precipitating a dramatic fall in data prices (over 90% within a year) and mass adoption (Krishnatri & Vellakkal, 2024). Between 2016 and 2021, wireless internet subscriptions more than doubled. By the end of 2021, over 96% of broadband users accessed the internet via mobile devices (Telecom Regulatory Authority of India, 2022). With mobile networks emerging as the primary channel for internet access in areas lacking wired broadband, the distribution of telecom infrastructure gained importance. Tower density, in particular, became a key determinant of connectivity.

Initially, network operators built and maintained their own towers, concentrating coverage in profitable urban centres. Subsequently, the diffusion of the “Towerco” model, in which independent firms build and operate towers for multiple service providers, reduced deployment costs and enabled scale-up in underserved rural areas (Azam et al., 2024). By 2020, a large majority of towers were managed by such independent firms, and the number of base stations expanded sharply from under 200,000 in 2008 to well into the millions by 2020 (Azam et al., 2024; Hounghonon et al., 2021). Reliance Jio’s rapid 4G expansion exemplified this shift, extending coverage into remote districts. Consequently, regional disparities in internet access became closely tied to district-level tower density. As such, it provides a credible source of variation for identifying the effect of internet access on women’s financial autonomy. Therefore, India’s transition from restricted, urban-centric connectivity to more mass, mobile-driven access, driven by policy initiatives, market disruptions, and innovations in infrastructure, creates a context in which internet access can translate into significant financial agency for women. Our analysis empirically quantifies that link.

Our findings indicate a substantial and statistically significant effect of internet access on women’s financial autonomy. In our preferred instrumental variable (IV) specification, access to the internet increases the likelihood of women exercising financial autonomy by 10 percentage points. This result remains robust across an extensive set of controls, alternative formulations of the treatment and outcome variables, and a series of falsification tests. The heterogeneity analysis shows that the effects are particularly pronounced for women with higher levels of education, and those from historically disadvan-

tagged social groups. We also explore mechanisms and provide suggestive evidence that internet access enhances financial autonomy by increasing women’s employment, improving their freedom of movement and financial awareness, which potentially improves their access to financial information and resources.

Our paper contributes to two strands of existing literature. First, we extend the literature on gender and financial inclusion by introducing women’s internet access as a novel determinant of financial autonomy. Demirgüç-Kunt et al. (2013) shows that women are systematically less likely than men to own bank accounts, save, or borrow, particularly in contexts where social norms and gender based legal discrimination limit their mobility, labour force participation, and control over assets. Prior work highlights the roles of individual characteristics (such as income, education, age, and employment), household factors (such as wealth and headship), and broader institutional conditions including banking costs, proximity to financial infrastructure, legal rights, corruption, and political stability in shaping access to and use of financial services (Dar & Ahmed, 2021; Ghosh & Vinod, 2017; Govindapuram et al., 2023; Kaur & Kapuria, 2020). While previous research has emphasized economic, demographic, and institutional correlates of financial inclusion, far less is known about how access to digital information and online platforms influences women’s financial decision-making and control.

Second, our study adds to the growing literature on how internet access drives societal change through diverse socioeconomic outcomes. Prior research shows that digital exposure enhances women’s well-being, improving maternal and reproductive health (Krishnatri & Vellakkal, 2024), delaying marriage (Murray, 2020; Zhang et al., 2023), and reducing intimate partner violence (IPV) (Dhamija et al., 2025). Internet use also improves labour market outcomes, benefiting women and the broader population (Bahia et al., 2024; Jain & Chatterjee, 2024), while fostering nutritional knowledge, dietary diversity (Deng et al., 2024; Vatsa et al., 2023), and civic participation with implications for political behaviour (Donati, 2023; Guriev et al., 2021). At the same time, digital engagement can exacerbate mental health challenges (Arenas-Arroyo et al., 2025; Golin, 2022). By situating women’s internet access within the context of financial autonomy, our study extends this literature into the domain of financial inclusion.

While prior research has emphasized the structural barriers to financial inclusion and the diverse socioeconomic effects of digital connectivity, little attention has been paid to how internet access directly shapes women’s financial autonomy. By bridging these literatures, we highlight the internet not merely as an infrastructure of communication, but as a transformative instrument that can reconfigure gendered power relations in financial decision-making. In doing so, our work underscores the importance of focussing on digital inclusion in gender-just policies, offering new insights into how technological change intersects with social norms and institutional contexts in shaping development outcomes.

2 Background

Within the broader domain of the financial ecosystem, a distinction exists between financial access and financial inclusion, even though they are often used interchangeably. Financial access, is broadly defined as “availability of a supply of reasonable quality financial services at reasonable costs, where reasonable quality and reasonable cost have to be defined relative to some objective standard, with costs

reflecting all pecuniary and nonpecuniary costs” (Claessens, 2006). In its simpler interpretation, it simply denotes the use of such products and services (Huang et al., 2016). However, as Demirgüç-Kunt and Singer (2017) emphasises, meaningful financial participation requires both access and actual utilisation. The World Bank defines financial inclusion as access to and the regular use of a range of financial products and services, including credit, savings, payments, and insurance, delivered in a responsible and sustainable manner (World Bank Group, 2023). This evolution in definition reflects a broader recognition that nominal access does not guarantee meaningful participation when services remain underutilised.

Financial inclusion is now widely recognised as a cornerstone of economic and sustainable development (Basnayake et al., 2024). Global trends over the past decade underscore this momentum. According to Klapper et al. (2025), the share of adults with a bank account surged from 51% in 2011 to 79% in 2024. This improvement is majorly attributed to the low- and middle-income economies, where account ownership grew from 42% in 2011 to 75% in 2024 with a striking increase of 33 percentage points. Much of this expansion has been driven by the rapid diffusion of mobile-based digital financial services. Yet significant disparities persist: women in low-income settings remain 6 percentage points less likely than men to hold an account (Demirgüç-Kunt et al., 2022).

India provides an instructive example of a large economy attempting to close these gaps. Over the last two decades, the Reserve Bank of India (RBI) has introduced a series of reforms to raise financial penetration, expand digital service delivery, improve credit access, and widen insurance coverage. The Pradhan Mantri Jan-Dhan Yojana (PMJDY), launched in 2014, marked a turning point by offering zero-balance accounts to previously unbanked households (Chowhan & Pande, 2014). A decade later, more than 56 crore accounts have been opened, with women holding 55.7% of them (The Economic Times, 2025). Account ownership among adults increased from 53% in 2014 to 89% in 2024 (Klapper et al., 2025), and many of these accounts have become active conduits for government transfers (Ajayan, 2023).

Alongside account ownership, India’s digital payments architecture has further reshaped financial inclusion. The UPI, launched in 2016 by the National Payments Corporation of India, consolidated multiple bank accounts into a single mobile platform, enabling frictionless transfers and expanding the reach of digital finance (Ministry of Finance, 2024). By late 2021, UPI was processing over 3.65 billion monthly transactions valued at Rs 7.71 trillion (The Indian Express, 2021). Initiatives such as “Unified Payments Interface (UPI) for Her” have specifically sought to deepen women’s participation in digital finance (Basu Chandola, 2025).

3 Data

We base our analysis on the fifth round of the National Family Health Survey (NFHS-5), conducted in India during 2019–21 (IIPS & ICF, 2021). NFHS is part of the global Demographic and Health Survey (DHS) program and provides nationally representative, cross-sectional data on demographic, health, and socioeconomic outcomes.¹ In India, the survey is implemented by the International Institute for Population Sciences (IIPS), Mumbai, under the supervision of the Ministry of Health and Family Welfare. Sampling follows a stratified random design, and the women’s questionnaire was administered

to all eligible respondents aged 15–49 across 636,669 households. The questionnaire covers fertility, contraception, nutrition, marriage, employment, autonomy, and domestic violence, with certain modules (for instance financial autonomy) fielded only in a randomly selected subsample of households.²

3.1 Financial Autonomy

Our main outcome variable captures women’s control over financial resources. We construct a binary indicator using responses to two questions: (i) whether the respondent has any money that she alone can decide how to use, and (ii) whether she has a bank or savings account that she operates herself. We code the outcome as 1 if either of these conditions is satisfied and 0 otherwise (Shirisha & Anjali, 2022). This measure reflects both the ability to exercise decision-making authority over personal financial resources and the utilization of financial instrument.

3.2 Variable of Interest and Other Controls

The primary variable of interest in our analysis is women’s access to the internet, coded as 1 if a respondent reports having ever used the internet and 0 if she has not. To address omitted variable concerns when estimating the causal effect of internet exposure on women’s financial autonomy, we incorporate an extensive set of controls. These include characteristics at the individual, household, and district levels. Individual characteristics include woman’s age (in years), completed years of education, marital status (yes/no/never in union), exposure to mass media (yes if exposed to newspaper, radio, or television), land or housing ownership (yes if sole or joint ownership), and duration of residence at the current location (in years).

Household-level controls comprise the gender and age of the household head, household size, religion (Hindu, Muslim, or other), social group (Scheduled Castes, Scheduled Tribes, Other Backward Classes, other social group/none, or don’t know/missing)³, wealth status (poor or non-poor), and area of residence (rural or urban). District-level socioeconomic characteristics, drawn from the 2011 District Census Abstract, include number of primary, middle, and secondary schools; medical, engineering, arts/science, and vocational colleges; family welfare centres; maternity and child welfare centres. We additionally include district-level sex ratio and SC/ST population share. The district-level count of commercial bank branches is available in the Reserve Bank of India’s Commercial Bank Directory (Garg & Gupta, 2023).

Geographic and infrastructural characteristics account for variation in physical terrain and local economic activity. Elevation is taken from the Shuttle Radar Topography Mission (SRTM); ruggedness is measured as elevation variation (Nunn & Puga, 2012; Riley et al., 1999); nighttime lights (a proxy for local economic development and urbanization (Dhamija et al., 2025)) come from Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band (DNB), flown jointly by National Aeronautics and Space Administration (NASA) and National Oceanic and Atmospheric Administration (NOAA) (Elvidge et al., 2017); average number of lightning-day are sourced from the India Meteorological Department’s Annual Disaster Weather Report, 1969–2019; and vegetation continuous fields reflect annual tree cover based in the form of the percentage of each pixel under forest cover. Data on vegetation continuous fields

is generated from a machine learning model based on a combination of images from NASA MODIS and samples from higher resolution satellites (Townshend et al., 2011). To reduce the influence of extreme values, elevation, ruggedness, and nighttime lights intensity variables are log-transformed. Nighttime lights and vegetation measures are lagged by one year relative to the NFHS-5 survey year to capture pre-exposure conditions.

District-level socioeconomic and geographic indicators are drawn through the Socioeconomic High-resolution Rural–Urban Geographic (SHRUG) database. SHRUG is an open-access database that documents multidimensional development across more than 600,000 villages and towns in India.⁴ Additional details on the SHRUG data are provided in Asher et al. (2021). We match NFHS-5 data to SHRUG indicators at the district level and obtain complete information for 692 districts.

3.3 Analytical Sample

The state module in individual recode file of NFHS-5 comprises 108,785 women between the ages of 15 and 49 years. Our analytical sample is constructed in two steps: (i) excluding 2,817 women identified as visitors, and (ii) restricting to respondents with complete information on the outcome, treatment, and controls. Missing values in years of residence at the current location (110 cases), age of household head (17 cases), and district-level indicators (2,184 cases) reduce the analytical sample to 103,657 observations. A schematic of the sample restrictions is provided in Figure A1.

3.4 Summary Statistics

First, we present summary statistics for woman- and household-level characteristics in Table 1. In our analytical sample, 85% of women have financial autonomy, where around 78% have bank/savings account that they can use themselves and 50% have control over their own money. The average respondent is 30.5 years old, has completed 7.4 years of completed education, and has lived in her current location for 15.9 years. Around one-third report internet use, 70% are currently married, 76% report some media exposure, and 47% own land or housing. At the household level, 83% are male-headed, with an average head age of 48.2 years and household size of 5.3 members. Around three-fourth of households identify as Hindu, 13% Muslim, and 12% with other faiths. Social group composition includes 19% SC, 19% ST, 38% OBC, and 18% other social categories. Approximately, 43% fall in the poorest or poor wealth category⁵ 75% reside in rural areas.

Next, Table 2 provides district-level averages. According to the 2011 Census, districts contained on average 1,629 primary schools, 743 middle schools, and 327 secondary schools, alongside 2 medical colleges, 6 engineering colleges, 15 arts and science colleges, 14 vocational training institutes, 107 family welfare centres, 64 maternity and child welfare centres, and 5 bank branches per district. SC and ST groups account about one-third of the district population, and the average sex ratio stood at 947 women for every 1000 men.

4 Empirical Framework

4.1 Estimation Strategy

We use the following empirical specification to investigate the effect of women’s exposure to the internet on a woman’s financial autonomy :

$$Y_i = \beta_0 + \beta_1 InternetExposure_i + \beta_2 \mathbb{X}_{ih} + \beta_3 \mathbb{G}_d + \beta_4 (\mathbb{S}_d \times t) + \theta_s + \epsilon_{ihdst} \quad (1)$$

where we define Y_i as a binary indicator of financial autonomy for woman i . $InternetExposure_i$ denotes woman i ’s exposure to the internet; $\mathbb{X}_i$ is the vector of woman- and household- level characteristics; $\mathbb{G}_i$ is the vector of geographic characteristics of her district of residence; $\mathbb{S}_d$ is the vector of other socioeconomic characteristics of the districts interacted with the survey year t ; θ_s are the state fixed effects and ϵ_{ihdst} is the idiosyncratic error term. The coefficient of interest, β_1 measures the effect of women’s internet exposure on their financial autonomy. Standard errors are clustered at the NFHS primary sampling unit (PSU) level to account for intra-cluster correlation, and state fixed effects are included to control for time-invariant state-level characteristics that may shape both digital access and women’s financial outcomes.

For β_1 to be interpreted causally, we must assume that, conditional on our control variables, there are no unobserved factors jointly influencing internet exposure and financial autonomy. However, this assumption is unlikely to hold in a simple OLS estimation. For instance, women who are more ambitious, intrinsically motivated, and financially literate, may be both inclined to use the internet and more capable of being more financial independent. Similarly, factors such as spousal attitudes towards women’s independence, intra-household communications, how progressive the household is, may simultaneously affect women’s internet use as well as their financial autonomy. Additionally, digital literacy programs and development policies may jointly affect internet connectivity as well as women’s financial inclusion. The presence of such unobserved traits in the error term would render the OLS estimates to be biased and inconsistent if they are correlated with internet exposure. As such, due to endogeneity concerns on account of omitted variables and measurement error in self-reported internet use, we employ an IV approach. In our case, IV strategy exploits exogenous variation in internet availability arising from density of mobile towers at the district level to estimate the effect of a woman’s exposure to the internet on her financial autonomy.

The tower density variable is derived from the OpenCellID dataset which is an open-access⁶ and partially crowd-sourced repository⁷ of cellular tower locations. This database provides the latitude and longitude of Base Transceiver Stations (BTS). We match these BTS coordinates to NFHS districts using the DHS district shapefile and construct *TowerDensity* as the number of BTS per square kilometre in each district. Following standard practice, we use the natural logarithm of *TowerDensity* + 1 to mitigate skewness and facilitate interpretation. Henceforth, we call our instrument variable as *Adjusted Tower Density*. In our analytical sample, the mean number of telecom towers per district is 3,123.5, and the average tower density is 3.1. Figure 1 maps the spatial pattern of tower density across districts. The

distribution of *Adjusted Tower Density* is presented in Figure 2 .

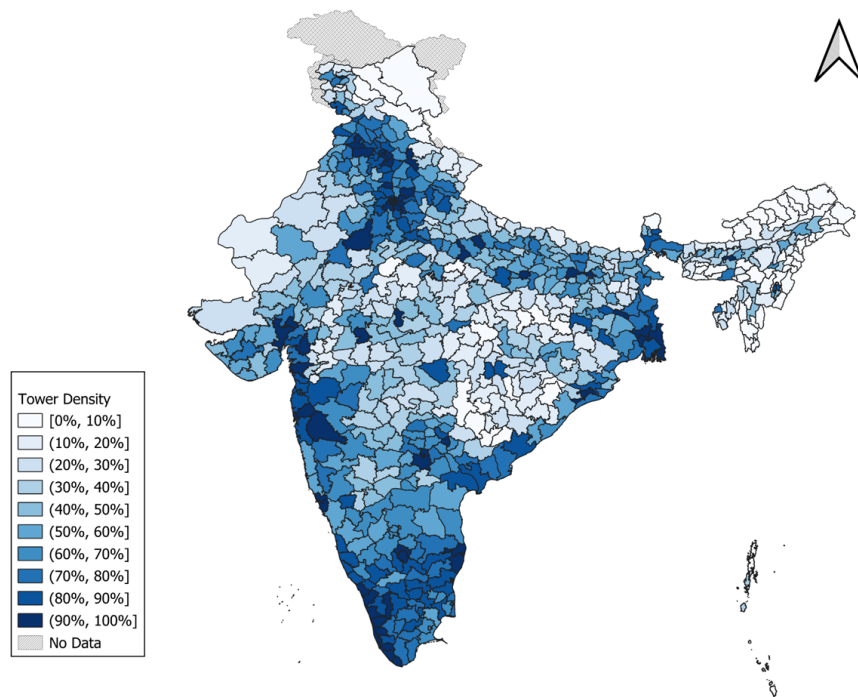


Figure 1: District-level map of Tower Density.

Notes: Plotted by authors using district coordinates from DHS as well as Survey of India and tower data is sourced from OPenCellID

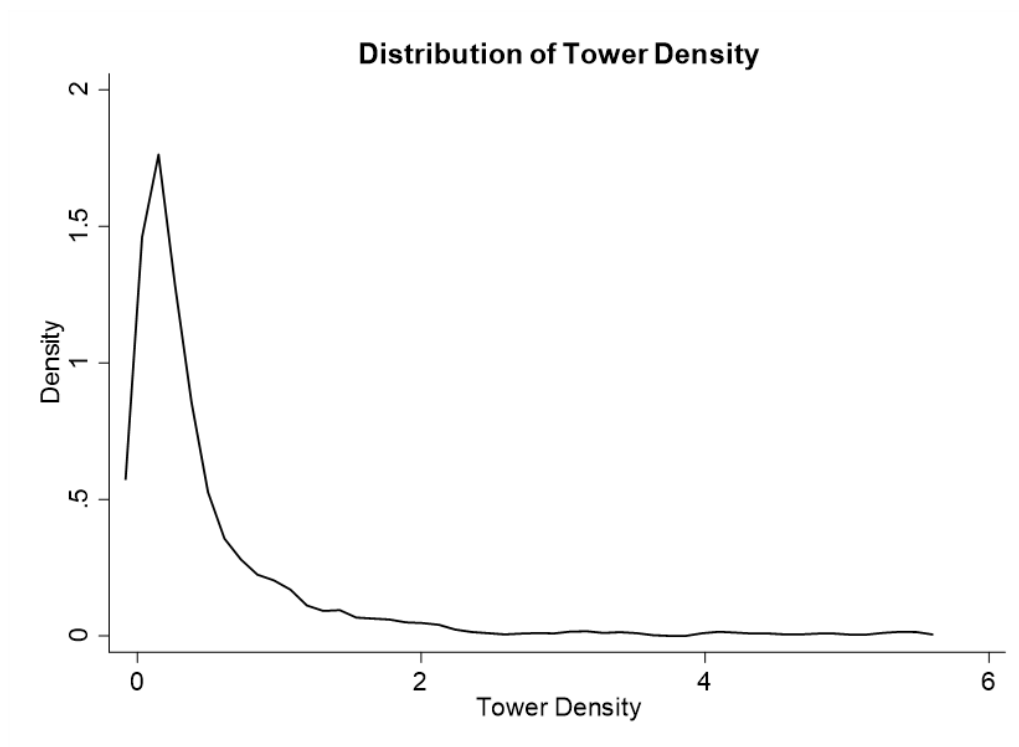


Figure 2: Kernel density plot of adjusted tower density.

4.2 Instrumental Variable Estimation

We estimate a two-stage IV model, which is specified as follows:

$$InternetExposure_i = \alpha_0 + \alpha_1 AdjustedTowerDensity_d + \alpha_2 \mathbb{X}_{ih} + \alpha_3 \mathbb{G}_d + \alpha_4 (\mathbb{S}_d \times t) + \phi_s + \eta_{ihdst} \quad (2)$$

$$Y_i = \beta_0 + \beta_1 InternetExposure_i + \beta_2 \mathbb{X}_{ih} + \beta_3 \mathbb{G}_d + \beta_4 (\mathbb{S}_d \times t) + \theta_s + \epsilon_{ihdst} \quad (3)$$

Equation 2 represents the first stage and Equation 3 captures the structural relationship. In the first stage, a woman's internet use, $InternetExposure_i$, is instrumented using district-level adjusted tower density. As in the Equation 1, $\mathbb{X}_{ih}$, $\mathbb{G}_d$, and $\mathbb{S}_d \times t$ are the vectors of woman/household level characteristics, district level geographic characteristics, and district level socioeconomic characteristics interacted with survey year t , respectively. ϕ_s or θ_s denote state fixed effects. Standard errors are clustered at the PSU level. Our parameter of interest, β_1 , captures the local average treatment effect (LATE) of women's internet exposure on their financial autonomy. Specifically, it identifies the effect for women whose internet exposure varies as a consequence of the instrument, namely $AdjustedTowerDensity$.

Given that both the treatment and outcome variables are binary, we follow earlier work (Addison & Teixeira, 2019; Babbar & Ojha, 2024; Chowdhury et al., 2018; Gershenson et al., 2022; Trinh, 2020) and estimate the system using the conditional mixed-processes (CMP) framework developed by Roodman (2011). CMP is well suited for models where the first stage involves a limited dependent variable and often delivers greater efficiency than linear IV approaches. In our setting, CMP functions as a limited-information maximum likelihood estimator, where only the coefficients from the structural equation carry the causal interpretation. The general likelihood function for this class of models is available in Roodman (2011).

4.3 Validity of the IV

This section evaluates the plausibility of using adjusted tower density as an instrument for women's internet exposure by examining whether it meets the relevance and exclusion requirements. Figure 3 shows a strong positive association between district-level averages of women's internet exposure and adjusted tower density. Next, we present our first stage estimates from Equation (2) in Table 3. We present the results in a sequential manner, starting with a specification with no controls (column 1), followed by controls at the woman and household levels (column 2), and then inclusion of district-level geographic and socioeconomic characteristics (column 3) and finally also including state fixed effects (column 4).

In all specifications, we use the log of tower density for ease of interpretation. Column (1) shows a strong and statistically significant association showing that a 1% increase in tower density is linked

to a 11 percentage point (pp) rise in the probability that a woman has ever used the internet. Across columns (2) and (4), while there is a fall in the magnitude of the coefficients, there remains a positive and statistically significant relationship between adjusted tower density and internet exposure. Column (4) presents our most preferred specification and we note that a 1% increase in adjusted tower density is associated with an 8 pp increase in the likelihood of internet use among women. Our instruments fare well on the diagnostic tests for IV. The Kleibergen-Paap rk LM statistic (54.27) and the first-stage F-statistic (55.13) provide support for the relevance of our instrument. Across all specifications, the F-statistic comfortably exceeds the conventional threshold of 10, helping rule out concerns around weak identification.

A second requirement for the validity of our empirical strategy is that adjusted tower density must influence women’s financial autonomy *only* through its effect on internet access. A potential threat arises if district-level economic conditions shape telecom infrastructure. When operators bear the cost of installing towers, network expansion tends to concentrate in commercially attractive areas. Potentially, it can create a strong correlation between tower density and local socioeconomic profiles (Azam et al., 2024). At the same time, these very district characteristics such as wealth, financial infrastructure, or gendered institutional norms, are often linked to women’s financial autonomy (Allen et al., 2016; Demirgüç-Kunt et al., 2013; Goodman, 2018; Omar & Inaba, 2020; Park, 2015). If unaccounted for, this would imply that adjusted tower density could affect financial autonomy directly, thereby violating the exclusion restriction.

To address this concern, we adopt two strategies. First, we saturate our IV model with an extensive set of controls capturing geographic, infrastructural, and socioeconomic features of districts, alongside state fixed effects, thereby limiting the scope for residual confounding. Second, we implement a residual-based diagnostic. We obtain residuals from regressing financial autonomy on internet exposure, and separately from regressing adjusted tower density on the same variable. Figure 4 plots these two sets of residuals. The absence of any visible pattern or systematic association provides a simple but informative check that, conditional on internet exposure, the instrument shows no direct relationship with financial autonomy.

Figure 3: First stage correlation between average district level internet exposure and district level adjusted tower density

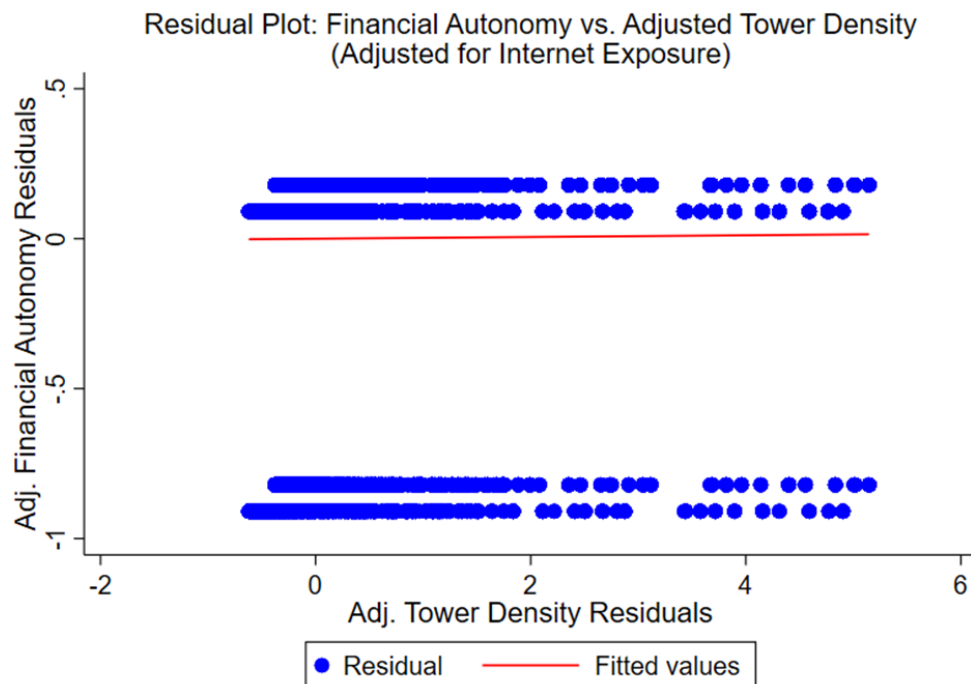
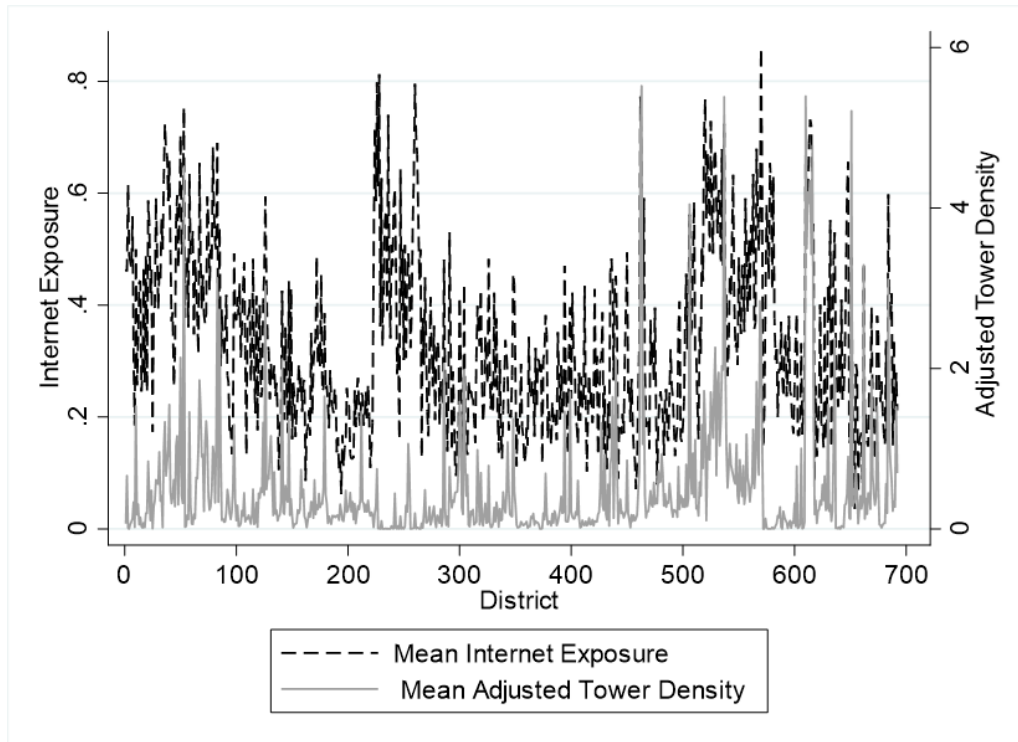


Figure 4: Residual plots: Financial Autonomy vs Adjusted Tower Density

5 Results

5.1 Baseline Results

We present the baseline results in Table 4. Given the binary nature of the outcome variable, whether or not a woman has financial autonomy, we estimate a Probit model in columns (1) to (4) of Panel A. Column (1) reports the unconditional association between women’s internet exposure and financial autonomy with no controls. We sequentially add controls for women’s and household characteristics, district-level geographic and socioeconomic factors, and finally, state fixed effects in columns (2) -(4), respectively. Across all specifications, the coefficient on internet exposure remains positive and statistically significant. In our preferred specification in column (4), which includes the full set of controls and state fixed effects, internet exposure is associated with a 10 pp higher probability that a woman enjoys financial autonomy.

Panel B reports the marginal effects from our instrumental variable (IV) estimation, following the same sequence of controls as in Panel A. Columns (5) - (8) progressively add individual, household, and district-level characteristics, and finally, state fixed effects. Across all specifications, the estimated effect of internet exposure remains positive, robust, and statistically significant. In our most preferred IV specification (column 8), we find that internet exposure increases the probability of women’s financial autonomy by 10 pp, and the effect is significant at the 1% level. This provides strong evidence of a causal effect of women’s digital exposure on their financial autonomy.

Notably, the magnitude and statistical significance of the estimated effects remain relatively stable across the Probit and IV-CMP specifications. This consistency suggests that potential endogeneity due to unobservables is unlikely to drive our results. In other words, while OLS and Probit estimates may overstate or understate the true relationship if internet exposure is endogenous, the similarity in magnitudes between the Probit and IV estimates indicates that such biases are likely minimal in our case. Our results highlight the critical role of digital connectivity for enhancing women’s financial agency. Internet exposure appears to equip women not only with financial awareness but also with the confidence and autonomy to utilize financial instruments effectively. As such, internet exposure can lead to greater economic inclusion and empowerment.

5.2 Components of Financial Autonomy

We also disaggregate our measure of financial autonomy into its two constituent components to examine whether the observed effects of internet exposure are driven by either or both dimensions. Specifically, we separately consider (i) whether the woman has any money of her own that she alone can decide how to use, and (ii) whether she has and uses a bank account in her own name. Both outcomes capture distinct aspects of women’s economic agency. The first reflects her direct control over liquid resources, and the second reflects inclusion in the formal financial system.

Using our IV specification, we find statistically significant and positive effects on both components. Table 5 presents the results. Column (1) shows that internet exposure increases the probability that a woman reports having her own money under her sole control by 14 pp. From column (2), we note

that the probability of her owning and using a bank account increases by 9 pp. These magnitudes are economically and statistically significant and suggest that digital connectivity not only facilitates access to financial instruments but also strengthens women’s individual control over financial resources. This is evidence that internet access enhances both the substantive and formal dimensions of women’s financial autonomy.

6 Sub-sample analysis

We now study the effects of internet exposure on women’s financial autonomy across several sub-samples. We consider sub-samples across women’s age, education, household wealth, social group, area of residence, and district-level banking infrastructure in Panel A and by regional classifications of Indian states in Panel B of Figure 5. Table A1 provides a detailed description of the construction of different subsamples. For each sub-sample, we re-estimate our most preferred specification including the full set of covariates and state fixed effects.

We find interesting heterogeneity in the effect of internet access on financial autonomy. In Panel A, across poor and non-poor households, in urban and rural areas, in districts with lower and higher availability of banking facilities, and for women above and below the median age, we consistently note statistically significant effects. However, when we examine the heterogeneity in impacts across women’s education, we find that the positive effect of internet exposure on financial autonomy is statistically significant only for the more educated women. The coefficient for the below-median education group is smaller and imprecisely estimated. This is consistent with a complementarity between digital exposure and human capital. More educated women are more likely able to convert online information into independent financial use on account of greater (digital) literacy and better labour market opportunities (Appaya & Abbas, 2024; Balasubramanian & Kuppusamy, 2020; Mishra et al., 2024). We also note statistically significant effects for women from the backward social groups, demonstrating a higher marginal return from internet exposure for the more constrained and disadvantaged women. In Panel B, we find statistically significant effects across all regions of India.

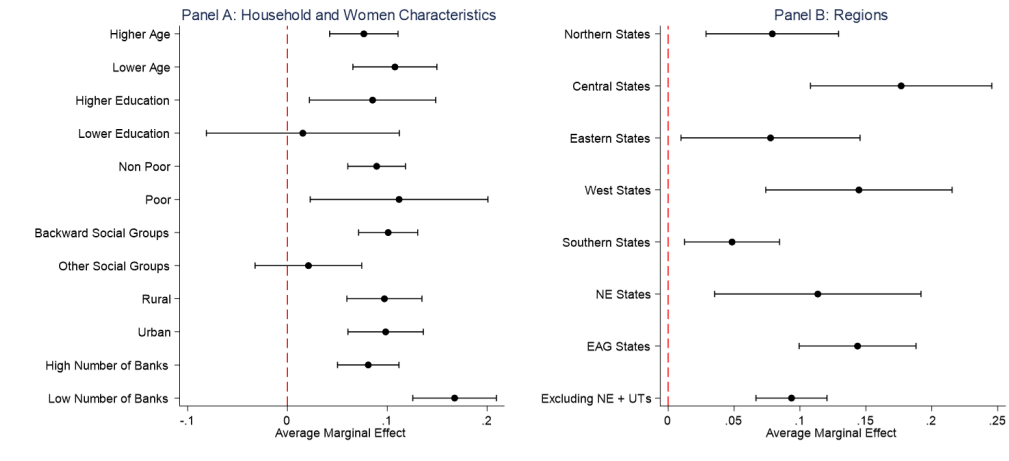


Figure 5: Heterogeneity analysis based on women’s characteristics, household characteristics, and regions.

7 Potential mechanism

We investigate three plausible channels to understand how internet exposure translates into greater financial autonomy. We consider women’s freedom of movement, their employment, and awareness about programmes focusing on financial instruments as the three mechanisms. Typically, these dimensions are consistent with mechanisms that link digital connectivity to economic empowerment. Women who are digitally well-connected have the potential to reshape social and gender norms through exposure to new ideas and opportunities (Chai et al., 2025; Zhou et al., 2020). Additionally, greater digital connectivity improves access to labour market information by reduction of information asymmetry and reduction in search costs (Jain & Chatterjee, 2024; Viollaz & Winkler, 2022). To study whether these act as channels of our estimated impacts, we adopt a two-step approach, similar to Mookerjee et al. (2023) and Bose et al. (2020), using the same source of exogenous variation in internet exposure, i.e. district-level adjusted mobile tower density.

In the first step, we examine women’s freedom of movement as a first potential pathway. The NFHS-5 asks women whether they are allowed to go to the market, visit a nearby health facility, and travel outside their village or community. We construct a binary indicator of mobility that takes the value one if the woman is permitted to go alone, and zero otherwise. This measure reflects the degree of autonomy in physical mobility within the household, which may itself be influenced by information access and shifts in social norms. The IV estimates (Panel A, Column 1, Table 6) indicate that an increase in internet exposure leads to an 17 pp increase in the women’s freedom of movement, significant at 1% level. In the second step, we examine whether mobility mediates the effect of internet exposure on financial autonomy. Specifically, we regress women’s financial autonomy on the predicted values of mobility obtained from the first-step. This specification isolates the exogenous component of mobility attributable to internet exposure. Standard errors are bootstrapped with 100 replications to correct for the generated-regressor problem inherent in two-step estimation. The results in Column (2) indicate a positive and significant association between exogenous improvements in women’s mobility and their financial autonomy.

Next, we estimate the effect of internet exposure on women’s employment, measured by a binary indicator equal to one if a woman reported being employed in the last 12 months. Panel B, Column (3) of Table 6 reports the first-step results for employment. Internet exposure leads to an 8 pp increase in the probability of employment. This is evidence that improved information access through internet exposure potentially increases women’s labour market opportunities and their participation in the workforce. In the second step, we regress financial autonomy on the predicted employment variable, again bootstrapping the standard errors. Results (Panel B, Column 4) show that predicted employment has a positive and statistically significant association with women’s financial autonomy, suggesting that labour-market participation is an important channel through which digital exposure enhances women’s financial decision-making capacity.

Finally, we examine whether internet exposure improves women’s awareness of programs that offer loans to women to start or expand their own businesses. Awareness is coded as a binary indicator equal to one if a woman reports knowing of any such programme. Panel C, Column (5) of Table 6 shows

that internet exposure, instrumented by adjusted tower density, raises awareness by about 6 pp. This indicates that expanded digital access improves information acquisition about credit schemes targeted at women. As above, we regress financial autonomy on the predicted awareness measure, bootstrapping standard errors. As shown in Panel B, Column (6), predicted awareness is positively and significantly associated with women’s financial autonomy, suggesting that improved financial awareness is a channel through which internet exposure strengthens women’s ability to exercise control over financial resources.

These results provide suggestive evidence that the estimated causal effect of internet exposure operates through these channels. By relaxing restrictions on women’s degree of freedom regarding mobility, expanding labour-market participation, and enhancing awareness about financial instruments, digital access relaxes both social and economic constraints that underpin women’s autonomy.

8 Robustness

8.1 Trimmed Instrumental Variable

Here, we test whether our baseline average treatment effects are driven by extreme values in mobile tower density at district level. To do so, we trim our instrumental variable, mobile tower density at district level, by top and bottom 1%, 5% and 10% of the distribution and re-run our baseline preferred specification. We present these results in Table 7, columns (1), (2) and (3), respectively. Our results remain robust across all three trimmed specifications. Column (1) records a causal effect of a 9 pp increase in women’s financial autonomy resulting from internet exposure. The coefficient remains stable in Column (2), and column (3) reveals an increase in women’s financial autonomy by 10 pp if the woman has internet exposure. These results present evidence that our baseline effects are not driven by outliers in the tower density across districts.

8.2 Additional Controls

Husband and Parental characteristics: Our baseline regressions include women-level characteristics, household characteristics, geographic factors, and socioeconomic indicators, along with state fixed effects. Given that decision-making within households is typically shaped by male authority, men often play a central role in determining women’s access to economic opportunities. Their influence, both before and after marriage, can significantly affect women’s ability and willingness to participate in formal financial systems. Thus, we augment our baseline model by including husband level characteristics such as husband’s age, years of education, employment status in the past 12 months along with woman’s exposure to inter-parental violence. We present the results in column (1) of Table 8. The marginal effect of internet exposure reduces marginally but remains statistically significant. Specifically, we note an increase in women’s autonomy by 6 pp due to internet exposure.

Neighbourhood infrastructure We may be concerned that the reported internet use might reflect a woman’s pre-marriage neighbouring district infrastructure rather than the district where she is surveyed, which we utilize to construct the IV. This would potentially violate the exclusion restriction if tower den-

sity in her district of origin is correlated with the instrument in her district of interview. To attenuate this concern, we augment the baseline with an additional covariate capturing spatial infrastructure, specifically, the average mobile-tower density in the neighbouring districts. Re-estimating with this control yield essentially unchanged coefficients. The internet exposure effect remains positive and statistically significant at approximately a 10 pp increase (Table 8, column 2).

Population density A remaining concern may be that densely populated areas receive better and more reliable digital infrastructure and financial markets compared to the sparsely populated districts. As such, the estimated relationship between internet exposure and financial autonomy may be confounded by settlement patterns. To address this, we augment our baseline specification to include district-level UN predicted population density for 2020 as an additional control. Column (3) in Table 8 shows that including population density leaves our estimated coefficient unchanged at 10 pp, showing that our baseline results remain robust.

8.3 Alternative Definition

Urbanization: We alter the measurement of the covariate capturing urbanization. In our baseline specification, we take nighttime lights intensity to serve as a proxy for the level of urbanization and local economic development. As an alternative measure, we replace nightlights with the district-level Multidimensional Poverty Index (MPI), as reported in the National Multidimensional Poverty Index Report (NITI Aayog, 2023), to capture a broader conception of development and deprivation at the district level. Table 8 column (4) presents the results. We continue to find a consistently robust and positive effect of women’s exposure to internet on her financial autonomy.

8.4 Falsification Test

Our main estimates are consistently positive and statistically significant for women’s financial autonomy across specifications so far. To rule out spurious correlation and strengthen identification, we add two randomization tests to the baseline as evidence of robustness of our results. First, we preserve the first stage relationship, that is, the correlation between the district level adjusted mobile tower density and women’s internet access, while randomly permuting the outcome of financial autonomy across women. More specifically, this randomization pairs woman i ’s internet access with another randomly chosen woman j ’s autonomy outcome. Second, we randomize the instrument by shuffling tower density across districts, keeping everything else unchanged. Doing so breaks the first-stage correlation. We run both tests of randomization 100 times each and plot the resulting t-statistic distributions in Fig. 6A, Fig.6B respectively.

From Fig 6A, we find that in 92% of the replications, there is no significant effect of internet exposure of woman i on another random woman j ’s financial autonomy. The second test which shuffles the instrument yields 95% insignificant associations between adjusted tower density and internet exposure 6B. Thus, these placebo tests show that our documented effect of internet exposure on financial autonomy is unlikely to be driven by random assignment of outcomes or by spurious correlations in the instrument.

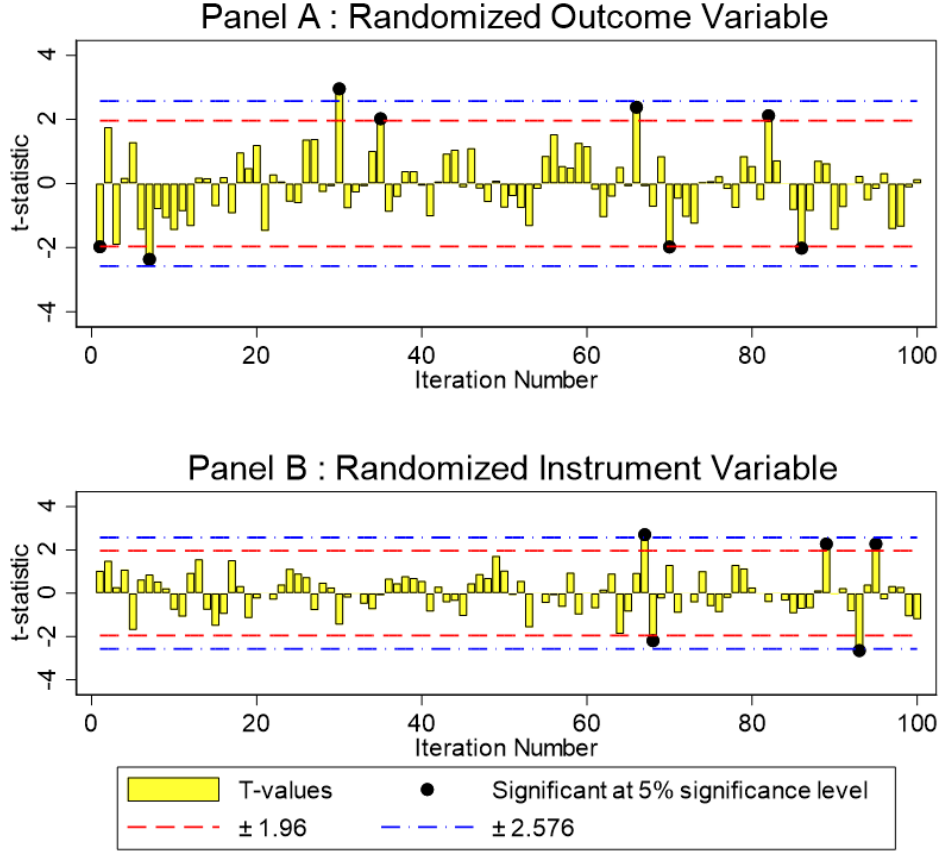


Figure 6: Falsification analysis

9 Conclusion

Our paper situates women’s financial autonomy within India’s entrenched patriarchal context that places women in low economic, social, and political hierarchies (Sarkar et al., 2025). We argue that a low degree of financial autonomy reflects limited financial and digital literacy (Kumar, 2022; Mishra et al., 2024; Ozili, 2025), time scarcity from unpaid care, mobility and safety constraints (Ozili, 2023), weak access to bank branches (Allen et al., 2016), documentation hurdles (Alliance for Financial & Inclusion, 2019), weaker institutions (gender-biased legal rights and political instability) (Allen et al., 2016), and unequal intra-household bargaining power (Duflo, 2012; Holloway et al., 2017; Sardinha et al., 2022). We also note that social restrictions and phone monitoring depress privacy and formal-service uptake (Barboni et al., 2018; Duvendack et al., 2023). Against this backdrop, our paper examines one determinant that can ease information constraints, reduce transaction costs, and shift bargaining frictions, viz., women’s exposure to the internet. We posit that internet use lowers search and transaction costs, expands private information and networks, and enables access to financial tools. Additionally, by raising earnings potential and mobility, it strengthens bargaining power and increases the likelihood that women control and use financial resources.

Using district-level mobile-tower density as an instrument for individual internet access, we estimate

the causal effect of digital connectedness on financial autonomy. We find that exposure to the internet raises women’s financial autonomy by about 10 percentage points. We then put our baseline through a battery of robustness checks and find the estimate to be stable across Probit and IV estimators, after trimming instrument outliers, and with additional husband and parental characteristics. We further show robustness to including a multidimensional poverty index (proxying development/urbanization) and UN-predicted 2020 population density, and we corroborate identification with randomization tests. Disaggregating the composite outcome, we find that internet exposure increases having money one can decide how to use by roughly 14 pp and personal use of a bank account by about 9 pp. We show that the effects remain consistently strong across sub-samples.

We also provide indicative evidence that improvements in women’s perceived mobility constraints, employment, and awareness act as plausible mechanisms of our effects. Internet exposure relaxes women’s mobility constraints by 17 pp, increases employment in the past year by 8 pp, and enhances financial awareness by 6 pp. This evidence is in line with the argument that digital access broadens women’s awareness and attitudes and job searches while also nudging their intra-household positions (Chai et al., 2025; Jain & Chatterjee, 2024; Li & Lu, 2024; Viollaz & Winkler, 2022; Zhou et al., 2020). Given that women’s financial empowerment carries wide welfare gains, with digital inclusion having the potential to generate downstream health and human-capital benefits (Sarkar et al., 2025), we assert that this question is of policy importance. Our findings indicate that extending reliable and affordable mobile internet in underserved districts can lead to important and tangible first-order benefits in women’s economic agency. There could also be further scope for policy prioritization where the impacts are stronger, such as among those more educated and those from the backward social groups who face greater structural constraints. While digital infrastructure is a necessary condition to generate larger and more durable effects on women’s agency, policymakers could consider bundling internet connectivity with targeted digital and financial literacy for women.

ENDNOTES

¹DHS datasets are available at <https://dhsprogram.com/>.

²This subsample is a state-specific module covering about 15% of the original households

³For brevity, we use the abbreviations SC (Scheduled Castes), ST (Scheduled Tribes), and OBC (Other Backward Classes) consistently throughout the manuscript.

⁴Available at <http://devdatalab.org/shrug>.

⁵The DHS wealth index is a standardized measure of household economic status that classifies households into five groups: poorest, poor, middle, richer, and richest.

⁶<https://datacatalog.worldbank.org/search/dataset/0038043/Global-OpenCellID-cell-tower-map>

⁷(<https://www.opencellid.org/>)

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Table 1: Descriptive Statistics: Woman- and household-level variables

Variable	Mean	SD	Min	Max
Outcome Variable				
Financial Autonomy				
Yes	0.85	0.36	0.00	1.00
No	0.15	0.36	0.00	1.00
Ownership of bank/saving account that they use themselves				
Yes	0.78	0.41	0.00	1.00
No	0.22	0.41	0.00	1.00
Control over own money				
Yes	0.50	0.50	0.00	1.00
No	0.50	0.50	0.00	1.00
Treatment Variable				
Internet use				
Yes	0.33	0.47	0.00	1.00
No	0.67	0.47	0.00	1.00
Women's Characteristics				
Age (in years)	30.54	9.91	15.00	49.00
Years of education	7.39	5.15	0.00	20.00
Currently married				
Yes	0.70	0.46	0.00	1.00
No	0.25	0.43	0.00	1.00
Never in union	0.04	0.20	0.00	1.00
Media exposure				
Yes	0.75	0.43	0.00	1.00
No	0.24	0.43	0.00	1.00
Land/House ownership				
Yes	0.47	0.50	0.00	1.00
No	0.53	0.50	0.00	1.00
Duration of residence (in years)	15.90	9.83	0.00	49.00
Household Characteristics				
Male Headed Household				
Yes	0.83	0.37	0.00	1.00
No	0.17	0.37	0.00	1.00
Age of HH head (in years)	48.22	12.84	14.00	95.00
Number of HH members	5.33	2.33	1.00	26.00
Religion				
Hindu	0.75	0.44	0.00	1.00
Muslim	0.13	0.33	0.00	1.00
Other	0.13	0.33	0.00	1.00
Social Group				
Scheduled Caste	0.19	0.39	0.00	1.00
Scheduled Tribe	0.19	0.39	0.00	1.00
OBC	0.38	0.49	0.00	1.00
Other/None	0.18	0.39	0.00	1.00
Don't Know/Missing	0.05	0.22	0.00	1.00
Wealth Category				
Poor	0.43	0.50	0.00	1.00
Non-poor	0.57	0.50	0.00	1.00
Residence				
Rural	0.75	0.43	0.00	1.00
Urban	0.25	0.43	0.00	1.00
Observations	1,03,657			

Notes: Women who are widowed, divorced, separated, or not living with their husband are coded as currently unmarried. Women have never married or whose gauna has not been performed are as categorised as 'never in union'. Women are considered exposed to mass media if they report exposure to any of the following: newspaper, radio, or television. Ownership of a house or agricultural land refers to either sole or joint ownership. Other religions include Christian, Sikh, Buddhist/Neo-Buddhist, Jain, Jewish, Parsi/Zoroastrian, no religion, and others. The bottom two quintiles of the DHS wealth index (poorest and poor) are coded as poor, while the top three quintiles (middle, richer, and richest) are coded as non-poor.

Table 2: Descriptive Statistics: District-level variables

Variable	Mean	SD	Min	Max
Socioeconomic Characteristics				
Primary schools	1628.62	1125.47	17.00	7097.00
Middle schools	742.75	549.05	0.00	4133.00
Secondary schools	326.79	301.85	1.00	2762.00
Medical colleges	2.29	4.75	0.00	51.00
Engineering colleges	6.36	12.74	0.00	148.00
Arts and science colleges	14.75	17.17	0.00	113.00
Vocational training institutes	14.23	24.10	0.00	269.00
Family welfare centres	107.13	1058.19	0.00	7196.00
Maternity and child welfare centres	63.87	105.03	0.00	1799.00
Number of Bank Branches	4.95	1.21	0.00	7.93
Sex ratio (female/male)	947.16	60.29	533.57	1184.40
SC and ST population share (in percentage)	33.09	22.74	0.34	98.58
Geographical Characteristics				
Elevation	5.49	1.21	1.41	8.53
Ruggedness	1.94	0.76	0.96	4.05
Nightlights	0.60	0.53	0.00	3.88
Number of lightning days	2.10	1.34	0.00	6.23
Vegetation Continuous Fields	16.84	15.13	0.00	72.89
Instrument Variable				
Number of towers	3123.51	7703.83	0.00	126593.00
Tower density	3.07	18.93	0.00	249.69
Adjusted tower density	0.49	0.77	0.00	5.52
Observations	692			

Notes: To minimize the influence of extreme values, the following variables number of bank branches, ruggedness, nightlights, elevation, and number of lightning days, are expressed in their natural logarithmic form after adding one to them. Similarly, adjusted tower density is computed as $\log(1 + \text{tower density})$. Tower density represents the number of towers within a district divided by its land area (measured in squares km).

Table 3: First stage analysis: Estimates of the relationship between adjusted tower density and women's internet exposure.

	Internet Exposure			
	(1)	(2)	(3)	(4)
Adjusted Tower Density	0.11*** (0.01)	0.03*** (0.00)	0.05*** (0.01)	0.08*** (0.01)
Observations	1,03,657	1,03,657	1,03,657	1,03,657
Women's characteristics	No	Yes	Yes	Yes
Household characteristics	No	Yes	Yes	Yes
Geographical characteristics	No	No	Yes	Yes
Socioeconomic Characteristics	No	No	Yes	Yes
State fixed effects	No	No	No	Yes
Kleibergen–Paap rk LM statistic	199.9	45.47	35.17	54.26
Kleibergen–Paap Wald rk F statistic	328.04	47.76	36.25	55.13
R-squared	0.05	0.33	0.34	0.35

Notes: Estimation via OLS regression analysis. Column (1) has no additional controls. Column (2) includes women- and household-level characteristics. Column (3) additionally adjusts for geographical and socioeconomic characteristics, while Column (4) further includes state fixed effects. Women's characteristics include age, years of completed education, current marital status, media exposure, duration of residence at the current location, and land/house ownership. Household characteristics include indicators for the sex of the household head, age of the household head, household size, religion, social group, wealth status, and area of residence. District-level socioeconomic characteristics include the sex ratio, percentage SC and ST population, number of primary schools, middle schools, secondary schools, medical colleges, engineering colleges, arts and science colleges, vocational colleges, family welfare centres, maternity , child welfare centres and number of bank branches. Geographical controls include district-level ruggedness, nightlights, elevation, annual number of days with lightning, and vegetation continuous fields. Standard errors (in parentheses) are clustered at the neighbourhood level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 4: Probit and IV analysis: Estimates of the effect of women's internet exposure on their financial autonomy.

	Women's Financial Autonomy							
	Panel A: Probit				Panel B : IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Internet Exposure	0.09*** (0.00)	0.09*** (0.00)	0.10*** (0.01)	0.09*** (0.00)	0.10*** (0.03)	0.09*** (0.01)	0.09*** (0.01)	0.10*** (0.01)
Observations	1,03,657	1,03,657	1,03,657	1,03,657	1,03,657	1,03,657	1,03,657	1,03,657
Women's characteristics	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Household characteristics	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Geographical characteristics	No	No	Yes	Yes	No	No	Yes	Yes
Socioeconomic characteristics	No	No	Yes	Yes	No	No	Yes	Yes
State fixed effects	No	No	No	Yes	No	No	No	Yes

Notes: Columns (1) - (4) presents the findings from the Probit estimation, whereas columns (5) - (8) reports the results from the Conditional Mixed Process (CMP) estimation. Columns (1) and (5) has no additional controls. Columns (2) and (6) include women- and household-level characteristics. Columns (3) and (7) additionally adjust for geographical and socioeconomic characteristics, while columns (4) and (8) further include state fixed effects. Women's characteristics include age, years of completed education, current marital status, media exposure, duration of residence at the current location, and land/house ownership. Household characteristics include indicators for the sex of the household head, age of the household head, household size, religion, social group, wealth status, and area of residence. District-level socioeconomic characteristics include the sex ratio, percentage SC and ST population, number of primary schools, middle schools, secondary schools, medical colleges, engineering colleges, arts and science colleges, vocational colleges, family welfare centres, maternity, child welfare centres and number of bank branches. Geographical controls include district-level ruggedness, nightlights, elevation, annual number of days with lightning, and vegetation continuous fields. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 5: Disaggregated outcome variable: Estimates of the effect of women’s internet exposure on specific indicators of women’s financial autonomy

	Women’s Financial Autonomy	
	Own money that one decides how to spend	Has own bank/savings account which they themselves use
	(1)	(2)
Internet Exposure	0.09*** (0.03)	0.14*** (0.02)
Observations	1,03,657	1,03,657
Women’s Characteristics	Yes	Yes
Household Characteristics	Yes	Yes
Geographical Characteristics	Yes	Yes
Socioeconomic Characteristics	Yes	Yes
State Fixed Effects	Yes	Yes

Notes: Estimation based on CMP analysis. The outcome variables are the variables which make up woman’s financial autonomy. Women’s characteristics include age, years of completed education, current marital status, media exposure, duration of residence at the current location, and land/house ownership. Household characteristics include indicators for the sex of the household head, age of the household head, household size, religion, social group, wealth status, and area of residence. District-level socioeconomic characteristics include the sex ratio, percentage SC and ST population, number of primary schools, middle schools, secondary schools, medical colleges, engineering colleges, arts and science colleges, vocational colleges, family welfare centres, maternity , child welfare centres and number of bank branches. Geographical controls include district-level ruggedness, nightlights, elevation, annual number of days with lightning, and vegetation continuous fields. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 6: Mechanism: Estimates of the effect of women's internet exposure on their mobility , employment and awareness.

	Panel A		Panel B		Panel C	
	IV	OLS	IV	OLS	IV	OLS
	Mobility	Financial Autonomy	Employment	Financial Autonomy	Awareness	Financial Autonomy
	(1)	(2)	(3)	(4)	(5)	(6)
Internet Exposure	0.17*** (0.02)		0.08*** (0.03)		0.06** (0.02)	
Fitted Mobility		0.54*** (0.02)				
Fitted Employment				0.82*** (0.03)		
Fitted Awareness						1.28*** (0.05)
Observations	1,03,657	1,03,657	1,03,657	1,03,657	1,03,657	1,03,657
Women's Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Socioeconomic Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Columns (1), (3) and (5) report estimates from Conditional Mixed Process (CMP), whereas columns (2), (4) and (6) present results from OLS regression. All the three panels include women- and household-level characteristics, geographical characteristics, socioeconomic characteristics, and state fixed effects. Women's characteristics include age, years of completed education, current marital status, media exposure, duration of residence at the current location, and land/house ownership. Household characteristics include indicators for the sex of the household head, age of the household head, household size, religion, social group, wealth status, and area of residence. District-level socioeconomic characteristics include the sex ratio, percentage SC and ST population, number of primary schools, middle schools, secondary schools, medical colleges, engineering colleges, arts and science colleges, vocational colleges, family welfare centres, maternity , child welfare centres and number of bank branches. Geographical controls include district-level ruggedness, nightlights, elevation, annual number of days with lightning, and vegetation continuous fields. Standard errors (in parentheses) are clustered at the cluster level in columns (1), (3) and (5). Standard errors are bootstrapped with 100 replications to avoid the generated regressor problem in columns (2), (4) and (6). ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 7: Robustness analysis by trimming the extreme values of the instrument variable.

	Women's Financial Autonomy		
	(1)	(2)	(3)
Internet Exposure	0.09*** (0.01)	0.09*** (0.01)	0.10*** (0.02)
Observations	1,01,247	93,189	82,808
Women's Characteristics	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes
Geographical Characteristics	Yes	Yes	Yes
Socioeconomic Characteristics	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes

Notes: Estimation via Conditional Mixed Process (CMP) estimation. Columns (1), (2), and (3) excludes the top as well as bottom 1%, 5%, and 10% of the adjusted tower density, respectively. All specifications include women's characteristics, household characteristics, district-level socioeconomic and geographic controls, and state fixed effects. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Table 8: Robustness analysis by including additional and alternative controls.

	Women's Financial Autonomy			
	(1)	(2)	(3)	(4)
Internet Exposure	0.06*** (0.02)	0.10*** (0.01)	0.10*** (0.01)	0.10*** (0.01)
Observations	57,823	1,03,657	1,03,657	1,03,657
Women's Characteristics	Yes	Yes	Yes	Yes
Household Characteristics	Yes	Yes	Yes	Yes
Geographical characteristics	Yes	Yes	Yes	Yes
Socioeconomic Characteristics	Yes	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes	Yes
Husband & Parental Characteristics	Yes	No	No	No
Neighbouring Districts' Tower Density	No	Yes	No	No
Population Density	No	No	Yes	No

Notes: Estimation via Conditional Mixed Process (CMP) estimation. All specifications include women's characteristics, household characteristics, district-level socioeconomic and geographic controls, and state fixed effects. Additionally, column (1) includes husband and parental characteristics: husband's age, education, employment in the last 12 months, and woman's exposure to interparental violence (=1 if yes and 0 otherwise). Column (2) includes the average tower density of all the neighbouring districts as an additional control. In column (3), we have added UN predicted population density for the year 2020 as a control. In column (4) we replace nightlights with multidimensional poverty index as an alternative control for urbanisation. Standard errors (in parentheses) are clustered at the cluster level. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Appendix

Figure A1: Analytical Sample formation

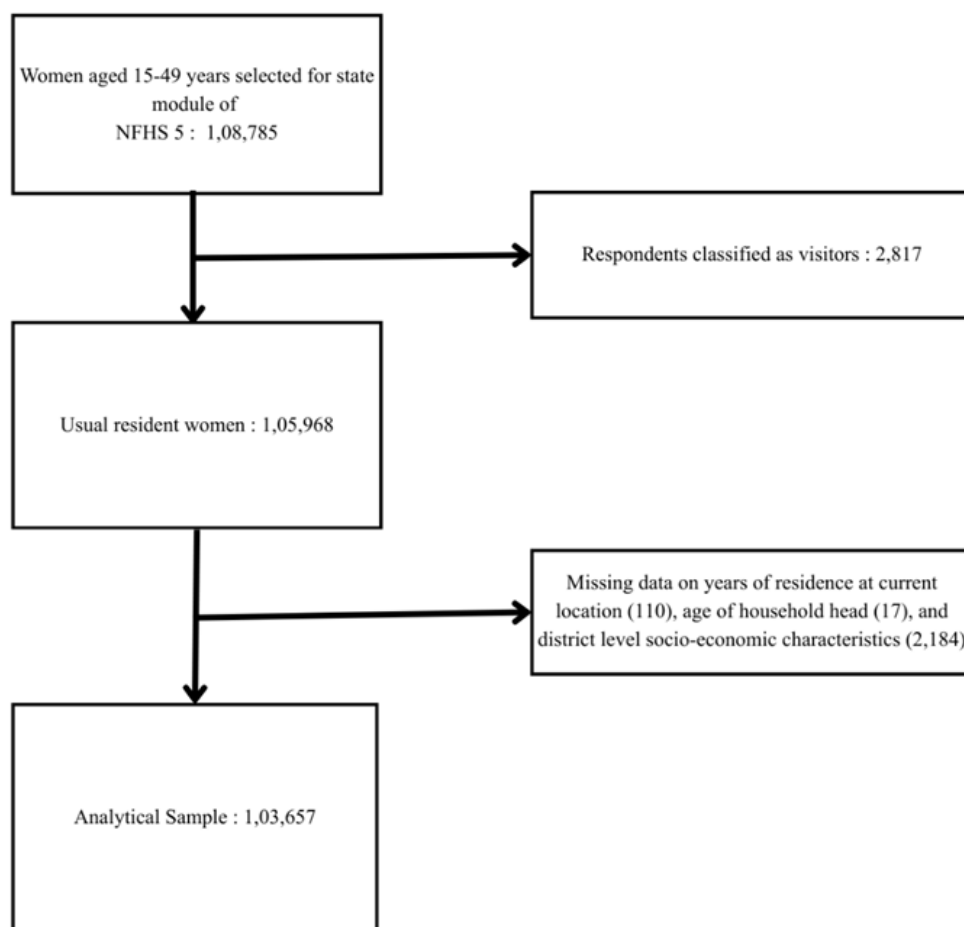


Table A1: Details of Subsamples

Characteristic	Description
Age	<i>Lower Age</i> group consists of woman aged 15–29, whereas <i>Higher Age</i> group included woman aged 30–49.
Education	<i>Lower Education</i> group included women whose years of education are fewer than 8; <i>Higher Education</i> group included woman who have 8 or more years of education.
Household wealth	<i>Poor</i> subsample consists of woman who comes from poorest or poorer households and the <i>Non-poor</i> consists of women who belong to middle, richer, or richest households.
Social Group	Women belonging to SC/ST/OBC backgrounds are classified as <i>Backward Social Group</i> ; women from Other Castes are classified as <i>Other Social Group</i> .
Area of Residence	Subsamples are created based on women residing in <i>rural</i> households and women residing in <i>urban</i> households.
Number of Bank Branches	Women are classified as living in a district with a <i>High Number of Bank Branches</i> if their district of current residence has number of commercial bank branches greater than or equal to the median (163); otherwise, they are classified as living in a district with a <i>Low Number of Bank Branches</i> .

Region	Description
Northern Region	Jammu and Kashmir, Himachal Pradesh, Punjab, Chandigarh, Uttarakhand, Haryana, Delhi, and Rajasthan.
Central Region	Uttar Pradesh, Madhya Pradesh, Chhattisgarh
Eastern Region	Bihar, West Bengal, Jharkhand, Odisha
Western Region	Gujarat, Dadra and Nagar Haveli and Daman and Diu, Maharashtra, Goa
Southern Region	Andhra Pradesh, Telangana, Karnataka, Lakshadweep, Kerala, Tamil Nadu, Puducherry, Andaman & Nicobar Islands
Northeastern Region	Sikkim, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Tripura, Meghalaya, Assam
Empowered Action Group (EAG) States	Bihar, Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh, Uttar Pradesh, Uttarakhand, Rajasthan
Excluding Northeastern States and Union Territories	All states listed above except Northeastern region and Union Territories are removed from the sample